

DameLibraries

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Why DameLibraries?

- ▶ This speech is about a methodology
- ▶ To reach good practiques when you are learning python libraries
- ▶ Programming oriented to tests is a good pratique
- ▶ Publish code packaged in a standard way is a good pratique

Tests Philosophy

- ▶ You must write your own test/s for your method before you write the method.
- ▶ You must not copy and paste code. You must write your own code
- ▶ If you need learn a library, you can write tests.

Executing Tests (I)

```
$ pip3 install damenumpy
$ cd /usr/local/lib/python3.5/dist-packages/damenumpy
$ nosetests3 tests
$ nosetests3 tests/test_basics.py
$ nosetests3 tests/test_basics.py:TestBasics.test_indexing
```

Executing Tests (II)

```
$ pip3 install dameformats  
$ cd /usr/local/lib/python3.5/dist-packages/dameformats  
$ pytest-3 tests  
$ pytest-3 tests/test_damejson.py
```

Publishing Code philosophy

If you publish code:

- ▶ You can save your code in multiple computers in a comfortable way. Save money! No desgracias!
- ▶ Another people can see your code. To be recognized!
- ▶ You can obtain improvements to your code. Peer revision.
- ▶ Public code for public people. To be popular

Executing publishing code

- ▶ Github/Gitlab is your social network. To be clever.
- ▶ Savannah is your social network. You are a GNU.
- ▶ Pipy is your social network. To be a popular Python programmer.

Create a repository in github

You can create a repository from the web interface and:

```
echo "# deleteme" >> README.md
git init
git add README.md
git commit -m "first commit"
git remote add origin https://github.com/davidam/deleteme.git
git push -u origin master
```

You can learn to create a repository in your server and learn to configure it in:

- ▶ <http://www.davidam.com/docu/crear-repositorio-git-servidor.html>

Packaging (I). Get a simple setup.py

```
$ git clone https://github.com/davidam/damenumpty.git  
$ cd damenumpty  
$ cat setup.py
```

Packaging (II). Write a simple setup.py

```
from setuptools import setup

setup(name='damenumpy',
      version='0.0.3',
      description='Learning Numpy from Tests by David Arroyo Menéndez',
      classifiers=[
        'Development Status :: 3 - Alpha',
        'License :: OSI Approved :: GNU General Public License v3 (GPLv3)',
        'Programming Language :: Python :: 3.6',
        'Topic :: Scientific/Engineering',
      ],
      keywords='numpy tests',
      url='http://github.com/davidam/damenumpy',
      author='David Arroyo Menéndez',
      author_email='davidam@gnu.org',
      license='GPLv3',
      packages=['damenumpy', 'damenumpy.tests'],
      package_dir={'damenumpy': 'damenumpy', 'damegender.tests': 'damegender.tests'},
      install_requires=[
        'markdown',
```

Packaging (III). Commands

```
# damefunniest
```

```
# https://python-packaging.readthedocs.io/en/latest/minimal.html
```

To install from local:

```
$ pip install -e .
```

To install create tar.gz in dist directory:

```
$ python3 setup.py register sdist
```

To upload to pypi:

```
$ twine upload dist/damefunniest-0.1.tar.gz
```

To install from Internet:

```
$ pip3 install damefunniest
```

Packaging (IV). You can download the new package.

```
$ pip3 install damenumpy
```

PEP 8

The best guide style for Python

- ▶ [https://www.python.org/dev/peps/pep-0008/
#a-foolish-consistency-is-the-hobgoblin-of-little-minds](https://www.python.org/dev/peps/pep-0008/#a-foolish-consistency-is-the-hobgoblin-of-little-minds)

Teach Yourself Programming in Ten Years (Peter Norvig)

► <http://norvig.com/21-days.html>

How to Become a Hacker

- ▶ `http://www.catb.org/~esr/faqs/hacker-howto.html`

Revenge of the Nerds

► <http://www.paulgraham.com/icad.html>

Free as in Freedom

▶ <https://www.oreilly.com/openbook/freedom/>

Google Philosophy (Ten things)

► <https://www.google.com/about/philosophy.html>

Agile Manifesto

▶ <https://agilemanifesto.org/principles.html>

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